Learning Hyperparameters via a Data-Emphasized Variational Objective

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BACKGROUND

Our generic probabilistic model assumes observed data $\{y_i\}_{i=1}^{N}$ are *i.i.d.* conditioned on parameters $\theta \in R^D$:

$$p_\eta(y_{1:N}, \theta) = p_\eta(\theta) \cdot \prod_{i=1}^{N} p_\eta(y_i|\theta). \tag{1}$$

This template is instantiated by specifying a concrete likelihood $p_\eta(y_i|\theta)$ and prior $p_\eta(\theta)$. The subscript indicates possible dependence on hyperparameters η.

We assume $D \gg N$ and $q_\psi(\theta) = \mathcal{N}(\theta|\bar{\theta}, \bar{\sigma}_q^2 I_D)$. We can tractably estimate $\psi = \{\bar{\theta}, \bar{\sigma}_q\}$ by maximizing the evidence lower bound (ELBO), defined for our model p and approximate posterior q as $J_{\text{ELBO}} :=$

$$\mathbb{E}_{q_\psi(\theta)}\left[\sum_{i=1}^{N} \log p_\eta(y_i|\theta)\right] - D_{\text{KL}}(q_\psi(\theta) \| p_\eta(\theta)). \tag{2}$$

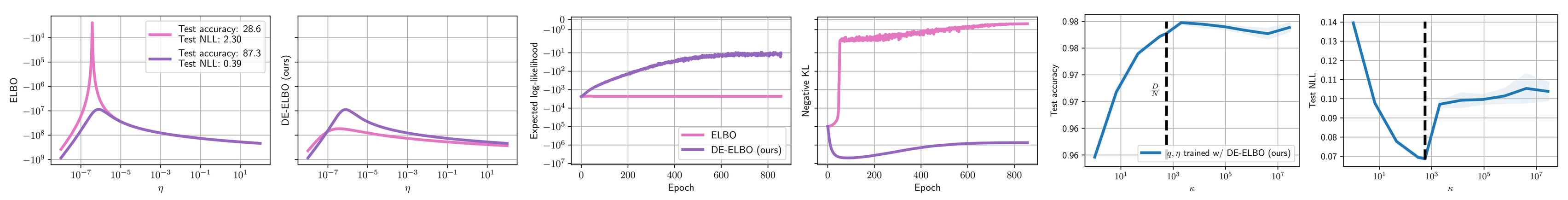
CONTRIBUTIONS

Contribution 1: When $D \gg N$ and q is a misspecified isotropic Gaussian, the ELBO prefers settings of ψ, η that underfit. Thus, gradient-based learning of both ψ, η via the ELBO yields notably subpar prediction compared to search methods for η .

Contribution 2: Emphasizing the data likelihood in the ELBO with upweighting factor $\kappa = \frac{D}{N}$ yields ψ, η values that fit data better when $D \gg N$ and q is misspecified. We propose the data-emphasized ELBO objective, $J_{\text{DE-ELBO}} :=$

$$\kappa \cdot \mathbb{E}_{q_\psi(\theta)}\left[\sum_{i=1}^{N} \log p_\eta(y_i|\theta)\right] - D_{\text{KL}}(q_\psi(\theta) \| p_\eta(\theta)) \tag{3}$$

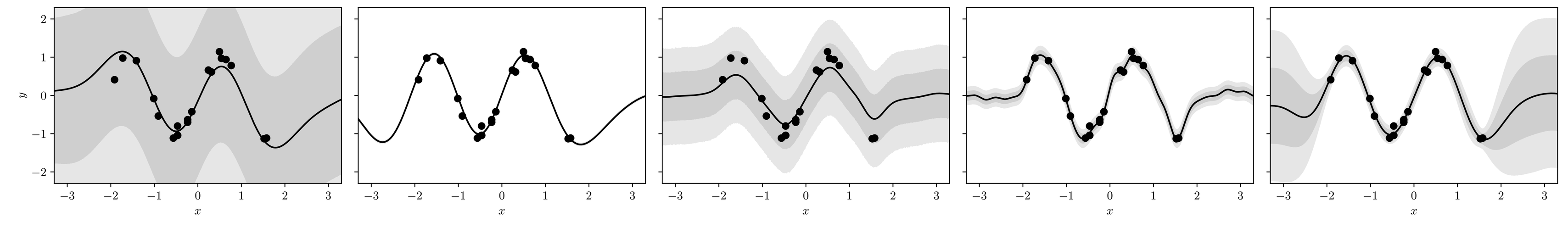
where we have introduced a scaling factor κ on the likelihood term. $\kappa = 1$ recovers the standard ELBO; instead, we recommend larger $\kappa = \frac{D}{N}$ to address the issues raised in Contribution 1. The DE-ELBO is equivalent to a standard ELBO for κN *i.i.d.* data instances, where we happen to observe κ copies of dataset $y_{1:N}$.



(a) Model selection as function of η (b) ELBO terms over training epochs (c) Performance as function of κ

Figure 1: *Panels (a)-(b):* Comparing approximate posteriors q trained for ELBO (pink) and DE-ELBO (ours, purple). **Takeaway: When $D \gg N$, the ELBO prefers simpler q (pink) close to the prior, while our DE-ELBO favors q with higher test accuracy (purple).** *Panel (c):* Test accuracy and negative log-likelihood (NLL, lower is better) for q trained via our DE-ELBO with various κ values. **Takeaway: Set $\kappa = \frac{D}{N}$.**

CASE STUDY A: REGRESSION



(a) diagEF LA-LML (b) diagEF LA-CLML (c) isotropic q ELBO (d) isotropic q DE-ELBO (ours) (e) $D \times D$ covariance LML

Figure 2: Predictions using ψ, η selected by different objectives for RFF regression. **Takeaway: DE-ELBO best approximates the true posterior's mean and variance near data. Variance far from data is underestimated.**

Model A Definition

$$\phi(x_i) = \sigma_k \sqrt{\tfrac{2}{R}} \cos\left(\tfrac{1}{\ell_k} A^\top x_i + b\right) \tag{4}$$

The non-learnable weights $A \in \mathbb{R}^{H \times R}$ and $b \in \mathbb{R}^R$ are randomly drawn *once* as $A_{h,r} \sim \mathcal{N}(0, 1)$ and $b_r \sim \text{Unif}(0, 2\pi)$ for all h and r.

Contribution: RFFs for arbitrary lengthscale. Our featurization in Eq. (4) generalizes the construction of RFFs that assumed lengthscale $\ell_k = 1$ and outputscale $\sigma_k = 1$ by Rahimi and Recht (2007) to any lengthscale $\ell_k > 0$ and outputscale $\sigma_k > 0$.

Point estimation view. $L(v) :=$

$$\sum_{i=1}^{N} \ell(y_i, v^\top \phi(x_i)) + \tfrac{1}{2}\|v\|_2^2. \tag{5}$$

Bayesian view.

$$p_\eta(v) = \mathcal{N}(v|0_R, I_R), \;\; p_\eta(y_i|v) = \mathcal{N}(y_i|v^\top \phi(x_i), \sigma_y^2) \tag{6}$$

Learning hyperparameters $\eta = \{\sigma_y, \ell_k, \sigma_k\}$.

Variational Methods for Model A

For simplicity, we chose a Gaussian with isotropic covariance

$$q_\psi(v) = \mathcal{N}(v|\bar{v}, \bar{\sigma}_q^2 I_R). \tag{7}$$

Theoretical Analysis of Model A

Lemma 1. Assuming isotropic q, as $D \to \infty$ the optimal approximate posterior variance $\bar{\sigma}_q^2$ for the ELBO ($\kappa = 1$) exactly matches the prior variance of 1.

Lemma 2. In the same setting as Lemma 1, as $D \to \infty$ the optimal approximate posterior variance for the DE-ELBO ($\kappa = \frac{D}{N}$) will be smaller than the prior variance.

CASE STUDY B: TRANSFER LEARNING

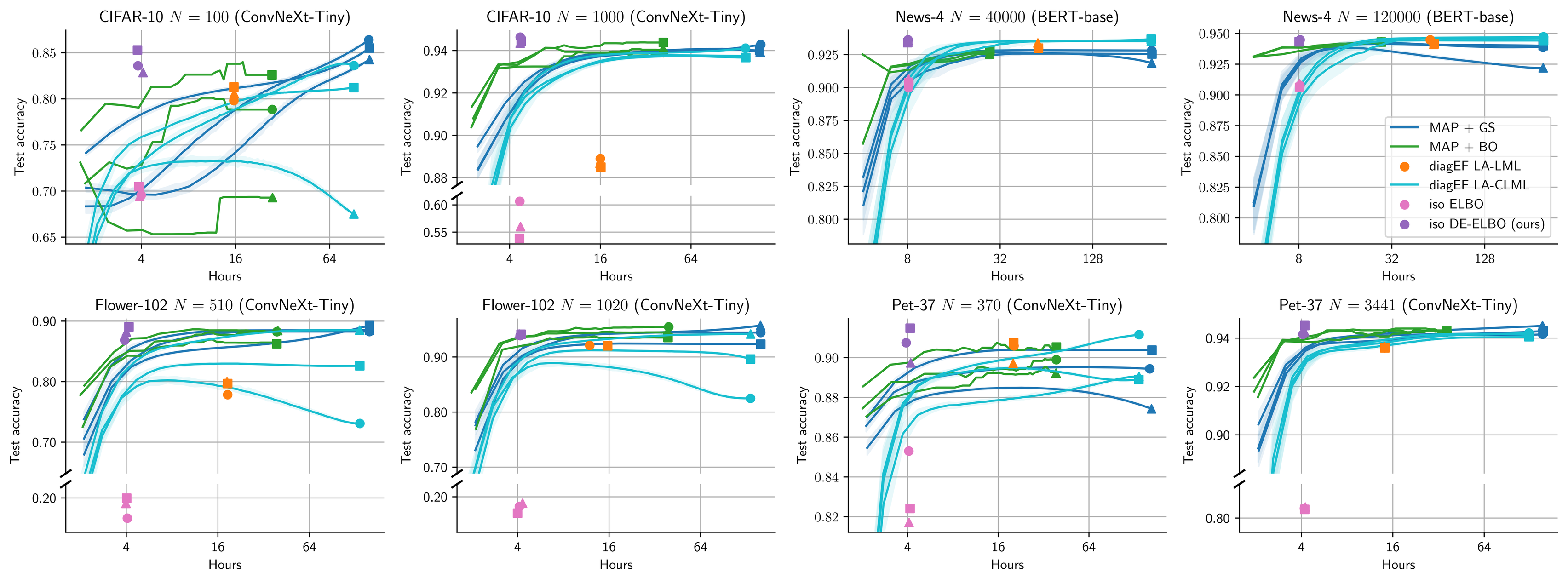


Figure 3: Test accuracy over time for L2-SP transfer learning of image and text classifiers. **Takeaway: After just a few hours, iso DE-ELBO reaches as good or better performance at small data sizes and similar performance at large sizes, even when other methods are given many additional hours.**

Table 2: Timing for transfer learning methods. **Takeaway: Each iso DE-ELBO run is 2x faster than diagEF LA-LML and learns λ, τ, avoiding the extreme costs of grid search.**

	Size of grid search (GS) space		L2-SP	
Method	λ, τ	Learning rate	**Avg. SGD run**	**Total GS time**
MAP + GS	36 (L2-SP) / 60 (PTYL)	4	37 min.	88.5 hr.
diagEF LA-LML	Learned via gradients	4	70 min.	4.7 hr.
iso DE-ELBO (ours)	Learned via Eq. (11)	4	33 min.	2.2 hr.

Model B Definition

Deep learning view. $L(w, V) :=$

$$\sum_{i=1}^{N} \ell(y_i, V f_w(x_i)) + \tfrac{\alpha}{2}\|w - \mu\|_2^2 + \tfrac{\beta}{2}\|\operatorname{vec}(V)\|_2^2. \tag{8}$$

Learning hyperparameters $\eta = \{\alpha, \beta\}$ (or equivalently $\eta = \{\lambda, \tau\}$).

Outlook. The DE-ELBO avoids the need for any validation set and expensive separate runs. It can offer hours of saved time to practitioners, which could be used to further improve models. Beyond saving valuable time, we hope our work sparks interest in theoretical understanding of modified ELBOs for improved model selection.

References

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Sanae Lotfi, Pavel Izmailov, Gregory Benton, Micah Goldblum, and Andrew Gordon Wilson. Bayesian Model Selection, the Marginal Likelihood, and Generalization. In *International Conference on Machine Learning (ICML)*, 2022.